

A Packing of 52 Lines in $\mathbb{R}^{15}$

Limin Ge*

October 8, 2026

Abstract

We give 52 1-dimensional subspaces of $\mathbb{R}^{15}$ whose smallest squared chordal distance is at least 0.935565. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.934976, obtained by deleting 1 of the 53 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 52 subspaces of dimension 1 in $\mathbb{R}^{15}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.935565\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 15$, $n = 1$. The cell $N = 52$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 0.934976, obtained by deleting 1 of the 53 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(15 - 1)/15 \cdot 52/51 = 0.951634$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m15n1N52.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 68 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
935565837871689370424066631667343426246744135845
denominator:
999999999999736387558560641715119981943905309374
```

B Coordinates

52 blocks of 15 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m15n1N52.txt`.

-262544346799	-128791792490	-258852534789	764515129410
-4854861760	200634351426	27021516206	238266312344
43808526802	-430240193911	92994847694	-45768848884
530504719235	324482747170	35471906288	-259643239557
-151898759068	-165470003155	-67781710616	-98564942415
-214830435320	30514653655	-270030418454	-108611672553
-10913837391	183821689738	589207600346	-127498910988
-208036370232	55632252878	-456775078418	159053199546
-321383236539	-135652376390	-66052973435	7509564913
-344816002245	-478780104168	-72414582666	-117545117924
-93203756064	-94035530676	190019757032	-133301693786
-69481809032	-97568542120	-257438182139	-62778306583
16051806399	-282067963388	408751263375	-228340940864
490906163377	415318416432	-171716410193	212296009295
-241085910423	248636273113	235165465925	305578238032
211465856518	190316220815	-169100995039	-401719437305
101193061063	308183748316	90299157864	-80357417472
103808098592	-374556444669	98834386798	-243485470137
-309099932068	34196738538	32630397995	-2918524908
-238735355560	331146621332	313342157038	-80171572700
273620493725	-191793506280	-143946309999	-96660351475
72848076638	-520315310998	184524988043	-399296911567
106265639251	145604398071	-664483655559	365214363077
172120391856	209801211557	52167018084	-150628608289
59880416019	-171117403908	30916803790	447912172739
-444601381406	-135797264404	359106306355	110635974950
603630465273	-569286064382	-203500572175	-19351524987
-39840624020	252364212033	202667666878	-330511428540
186766709029	50954899677	44742887852	-318014164735
-241952462797	-328271347100	-398432420417	134090723127
19073848404	16654205248	220609438047	-55413136139
94614308190	-137591614615	-230896678897	-136209898668
-365420782021	154186363686	-96635039257	-298498532651
-59282432762	-137184036877	-70674081560	-274552368759
-173172837161	-140048093929	153381788545	-17393235174
-160802944570	-287161163623	40173048491	128236746756
157813281089	65103092154	-383742973389	-337443501951
81361166166	498190479688	-406140399628	187938397921
-65602376296	-257757716582	-414817909788	214003234665
292438341173	511957949245	45478334555	352721299913
180926482048	-251047473335	-123993187837	-367667941979
522320759295	397596027022	495727986639	-163366447933
349589434167	-118761468930	-443845931253	445474692757
-387690467650	57585809466	-334569744328	10729231177
-313644038388	-137130109108	-21526703922	-342518458444
-298138435747	-32447172245	43192234902	488903799288
-340488934429	-215009504953	-284869749236	220904469259
-116724858747	38936701443	-201292737940	278602916688
-88724288259	-171375663249	133631466874	179764663505
134756893461	32652916628	-116796488651	640886566951
-234481687760	-86780574478	-454596217530	161757011908
-30406617018	169393695697	41335344657	-24857849748
16887306922	-206222661953	-180643851073	37041116222
736312851638	-127391721420	-139439517042	134877939524
-91673625484	270426154816	118339001835	201767716796
-223086986190	-236919844294	-59057114174	174914338438
167557063205	121185539347	523576956467	-231635751467
13415541577	-99288428136	299829786755	-192344206655
-109705025390	243938526551	320504693215	28752974778
242310193090	87369753767	155110988917	-137838230292
-384172745002	542784186511	-36461414401	-39037112006
-111774833858	33625831411	141481205269	245916906396
-36563651237	-479377412504	-116909329943	2438296338
-246226006215	-265315930008	-124893551623	341678561493
117736412923	-296871910790	133478939659	8251729698
119822551623	-159474857871	-409434186792	436037634456
230055565213	88847051394	183410118586	104632414636
-500691543217	-273004519361	255039564287	-162849497728
135894250695	412877996294	-166429409568	241615528754
-230286635903	33263282700	-378483966405	-13884170069

443035106508	349219790953	-360406240030	-189148247848
484050896859	-158742746087	-261235610314	71181281370
-6498357830	-2252017595	-115806127029	-114076270161
228169577390	-35413444963	-354782528911	-195213040259
-228749182711	231689828770	343707497457	730509305811
26210436115	-267975356906	-6112523418	-331156334867
132204459586	173838088914	-139754978067	16448318826
56524751828	134511200710	-473874767937	247707137215
-35988749529	-228563799916	189422210385	2693291064
-275419412771	-173747804859	222690019719	-105693200787
191011296449	384344594815	74244280960	65569361492
-21580437554	-43599350099	229783510967	-9909040144
-364426425098	380989764510	73221499254	-32803223075
584866811352	-16355825454	-73743384440	162727179573
-107666829523	-416146004946	-179418484278	95454288783
-64452393985	-122485897737	45515779368	81421402074
-31469045775	157141234412	-92136277850	268882098816
106221783689	105925003119	-383910774367	-117201356743
-431659117343	507768396552	318225106713	-6615825745
-418703770340	256129983731	-222338114137	-385878505368
37699230624	-36330695492	-132705360171	216751543735
557435108	1697466899	-222297773121	152812161326
377253957078	271440646822	9656460091	-58487789208
-231063557707	154503016698	-372014363665	31826132914
-473691070522	-224533588484	-198812447250	-229726107847
-290202842982	21371331623	-592045620770	515204766521
415009167951	239668801637	-241577699793	198794685614
-50338474927	-291778987738	48125564438	-196379274045
-177269910397	19054724223	201859703375	527464319113
106381314141	231632754966	-119696844577	-476042457049
-63144564734	-450153434871	-430455849253	270209240827
-196384164372	-92794260015	149286744193	381761219665
-81153720187	470876118916	59093102432	-110874125412
456590595473	-88592137262	209521813635	-258664604814
138360291123	126031869489	56468141844	-219986320569
-331790100047	-164016696904	-148294705042	-321978053871
178784167792	107861631336	647618678104	-71704678020
26979588112	-483138083301	15553643874	196879065945
-576046430867	235601309976	64318605912	153417671329
-227581993666	12212405404	-419851390359	-148180698895
49245118065	-21445334665	63524826349	-161774932450
181281895506	-6735755798	293327122888	379677416018
-123980824796	-277115118693	-6877974352	-152146533265
-243099590326	-111118069115	-463680088730	201227615131
293903828124	78430256577	169264482292	190112129347
-51670274725	485302209126	-249432101524	36666448821
-217641596294	-280470181737	-122456051871	-481521804993
-104219611569	-555429014296	142413259395	96026845902
-285445314014	135999464193	-64404621735	213749489367
-367540254477	331761952686	176390320505	495872834053
86385122647	238379495677	-296339371111	19963729069
386158787116	149102093717	45988778439	119321846418
173165440120	-135074331012	61388644187	-79341898793
197661782746	2371142922	-56742568445	134776799516
-365099607556	33581199792	-537702457342	-140272830235
372724800350	81798074506	389182604973	-427868496463
66682115696	236611647594	111454781580	60787250611
151408731114	40261564309	115572421322	23548858655
304467189185	422636297398	323162330840	428667822889
481654404927	431015498550	76743137290	-509818020451
-67152738713	420043841483	66487444057	106967963457
-307025339890	310524127604	-356689855085	42448758562
47459628457	-177714249443	927952555	-69383504154
211333193160	-140681124065	502699502328	42915209204
-66261025290	155142573039	376767526355	82126448454
-373520162295	362241248133	-147263689100	319976206042
223655628681	-289273881828	-365938733137	306695322419
24148784356	83492372470	-204704057181	15753930211
346172894554	114079437207	189987772054	257819280224
-135345388954	23127272838	273798142165	-599329348730
206730559666	-222874680695	-170368763452	-21735137367
-45709842214	-341347666087	222819257774	-168358258340
135782444728	200995263982	-153794266463	124885200723
-286391968735	388393380640	286906672790	216023939862
-398308669690	-373301463285	90513010070	30344683993
-233056292613	-152511061075	-298083578134	-264807097825
-210326500605	-285464293806	-182841629760	106863579086
-218149223015	-342285217958	230061911358	271106575333
-469123603842	55477898756	-438410707555	-493016803780
39638152736	128797949124	-408779512718	144551469523
120232559614	141844714753	-138799229599	228783223756
-167128060056	476476691161	-13205669625	201646208242
96641718130	30525981844	-102673812455	395624479272
-473751128427	-63358177846	358106533435	-316715712550
-114401633186	144578358481	-438419793470	-19208905015
390448879558	-42837558147	19775252084	3222574234
368333985717	-292029255694	-143982731227	-407602859327
4604839663	-236037215855	404660565754	194842507439
-273249130127	-304854378259	-258824401446	-160510176066
-414727383423	62059729312	442689416536	-261651366190
-205294185792	-137978272315	200832497314	-489671007424
1319610607	46229580288	-26316719985	-272036897254
-159814230822	-679703163084	296683896690	-59073079374
-170990219624	137235282889	-119239877721	41733013147
281093972947	-419358152323	477962947851	-440220961483
12347405109	-159261627403	-230110541474	159219374530
285131347569	-22769819259	-94657363251	288551083301
492614291819	-178091119158	-3042544439352	17100874657
268710995474	255534088729	-129914846065	117726236928
-235210893086	-269779795696	95834035049	-24003974462
-113680865	-171382637396	17228291075	-65749495362
-475533617171	204660672448	-387613278266	-181300959909

284217553929	185072343730	-62929455534	45072046847
-421209851426	-132128067795	-49512585687	169562259948
-2290371805	-152505641588	-72033769460	3985119864
337981286484	39798048225	-69725603104	-315837098531
-161064407354	-276148647973	135764723655	168059133001
412682903782	-67727912949	-54980456237	198725992344
-271648025183	-237331334075	-78029622654	-347694357598
-245340019976	-103604934041	-345497688508	-231679033661
-44877605271	317216398896	-621882149658	103702437579
-229786444068	-173024857673	85184371046	-265220163840
-328565100788	-167550110166	-1529756022	458259516316
348529630115	-250552602232	-177868209506	236795241363
-267511782912	-246616931613	523656198385	219998904496
-94912284116	-221151333321	367180073920	391047580139
-56416401803	451798088187	44606760070	-202159829227
-22763242523	405385279302	62656089730	-289317983476
432451728722	292497473605	561101010377	-281069481947
-131057431783	-181862306509	-507316216658	-268428618376
-694146411489	222491587795	82982660878	-100848170872
-98149905728	198725917163	45540285099	232057839273
404959225376	84596307752	98637755289	-170197058315
12663592569	221839198094	-178607028310	-102680409367
45999738488	364792740493	65967863458	-268986249036